

Vattanary Tevy

+1 310 438 8709 | vtevy@uw.edu | linkedin.com/in/vattanary-tevy | Seattle, WA 98117

SUMMARY

MS Mechanical Engineering researcher (UW, expected Jun 2026) with active research in imitation learning and reinforcement learning for contact-rich robotic manipulation. Leading a bimanual robotic composite layup system using ACT and Diffusion Policy with real-time impedance control on UR5e hardware, with a laser profilometer providing wrinkle detection and policy evaluation feedback toward DSRL. Holds the Amazon Robotics Research Fellowship. Background spans RL policy training, diffusion models (CIDER: 1D U-Net, MAE 0.0069 A), sim-to-real transfer (SHIELD: HJI+MPPI, >50% collision reduction), and production-scale statistical analysis at Amazon Robotics. Proficient in Python, PyTorch (CUDA), ROS2, MuJoCo, and LeRobot.

RESEARCH EXPERIENCE

WEIRD Lab — Washington Embodied Intelligence & Robotics Development Seattle, WA Jan 2026 – Present
Reinforcement Learning Researcher | Amazon Robotics Research Fellow

- Lead a bimanual robotic composite layup system using two UR5e arms: right arm executes ACT and Diffusion Policy (Chi et al., LeRobot) for contact-rich manipulation; left arm mounts a laser profilometer for wrinkle detection and policy evaluation — research in preparation for publication.
- Implement ACT and Diffusion Policy with real-time impedance control on hardware (ROS2, LeRobot); own full pipeline from teleoperation data collection through policy training, closed-loop evaluation, and benchmarking; building DSRL infrastructure using profilometer feedback as a learned reward signal.
- Design and fabricate end-effector subsystems and custom tooling in SolidWorks; validate through structured closed-loop hardware test cycles targeting stiffness, weight, and repeatability requirements.

uWAMIT – UW Molecular Imaging & Therapy Center
Mechatronics Researcher

Seattle, WA Jun 2024 – Mar 2025

- Built a custom photoacoustic-ultrasound scanner from concept to clinical-grade validated prototype (GE HealthCare): integrated optical, electronic, and mechanical subsystems; resolved laser-fiber synchronization failure at 3000 RPM through systematic fault isolation — met clinical image resolution targets; 1 co-authored publication.
- Built 1000 Hz multimodal real-time data acquisition system (fiber optics, laser timing, ultrasound) with custom Python and Automation1 control scripts.

Amazon Robotics

Seattle, WA Sep 2025 – Dec 2025

Robotics Data Scientist Intern | Amazon Robotics Research Fellow

- Confirmed a month-long unresolved production failure in a robotic vision system: applied chi-square and independence tests across millions of cycles — isolated root cause in under 2 weeks (>50% faster), directly unblocking the engineering fix.
- Designed cross-model consistency metrics for segmentation, depth, and dimension pipelines; built scalable Python pipelines over AWS S3 and SQL — production monitoring framework for threshold tuning and alerting.

SELECTED PROJECTS

SHIELD: Safe Real-Time Robot Controller

Seattle, WA Mar – Jun 2025

- Integrated HJI reachability into MPPI as a hard safety constraint: value function solved offline, enforced online — >50% collision risk reduction at 20 Hz CPU execution with only 4 ms overhead.

CIDER: Conditional Diffusion Model — Trajectory Generation

Seattle, WA Mar – Jun 2025

- Co-developed 1D U-Net diffusion model (PyTorch, CUDA) on 12,000+ simulated trajectories — MAE of 0.0069 A validated against real-world battery data.

3–6 DoF Robotic Arm: Kinematics & Motion Planning

Seattle, WA Mar – Jun 2024

- Designed 3–6 DoF arm in SolidWorks (URDF); implemented FK/IK and motion planning under ROS; validated in MuJoCo simulation.

PUBLICATION

*R. Shang, P. Lorraine, S. Smith, H. Chan, *V. Tevy, I. Pelivanov, M. O'Donnell. "Real-time spectroscopic PAUS imaging integrated on a commercial scanner." SPIE Photonics West, Jan 2025. * = equal contribution

Bimanual robotic composite layup with learned manipulation policies and profilometric quality evaluation. (in preparation).

EDUCATION

University of Washington

Seattle, WA Expected Jun 2026

M.S. in Mechanical Engineering – Data Science & Mechatronics

Graduate Coursework: Reinforcement Learning for Robotics, Deep Learning for Computer Vision, Convex Optimization, Data-Driven Dynamical Systems, Control Systems

University of Washington

Seattle, WA Jun 2024

B.S. in Mechanical Engineering – Mechatronics & Applied Mathematics

TECHNICAL SKILLS

Robot Learning: ACT, Diffusion Policy (Chi et al.), imitation learning, DSRL, reward design, real-time impedance control, sim-to-real transfer, VLA/policy evaluation, Hugging Face LeRobot, ROS2, MuJoCo

ML Engineering: PyTorch (CUDA), Python, policy gradients (PPO/TRPO), diffusion models, MPPI/MPC, HJI reachability, experiment design, benchmarking

Sensing & Hardware: Laser profilometry, force/contact sensing, multimodal sensor integration, UR5e arms, DAQ systems, embedded systems (ESP32, Embedded C), UART/I2C/SPI

Software & Data: Python (NumPy, SciPy, Pandas), SQL, AWS S3, Git, MATLAB/Simulink, Linux, C++ (familiar)

HONORS & FELLOWSHIP

- Amazon Robotics Research Fellowship, University of Washington (2024–2025)
- The Social Justice Award (Ethics), University of Washington (Jun 2024)